

Coding & Solution

Date	30 June 2026
Team ID	SWTID-2026-9325
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/mohan2005-code/DEMO/blob/main/README.md
Programming Language(s)	Python, JavaScript, HTML, CSS
Framework(s) Used	[e.g., Flask, React, Spring Boot]
Key Features Implemented	• User Authentication (Register/Login) • Financial Profile Management • Loan Management (Add/View/Delete Loans) • AI-Based Debt Stress Analysis • AI Negotiation Letter Generation using Google Gemini API • Repayment Simulation (Avalanche & Snowball Methods) • Financial Dashboard & Negotiation History
Pending / Incomplete Features	• Email Notification Integration (Optional) • Cloud Deployment • Advanced Analytics & Reports
Setup / Run Instructions	1. Install Python 3.13+ and Node.js. 2. Install backend dependencies using pip install -r requirements.txt. 3. Configure the Google Gemini API key in the environment file. 4. Start the FastAPI backend using uvicorn main:app --reload. 5. Install frontend dependencies using npm install. 6. Start the React frontend using npm run dev. 7. Open the application in the browser and log in as a borrower.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project follows a modular architecture using FastAPI, React.js, SQLAlchemy ORM, and SQLite. The backend is organized into separate modules for authentication, CRUD operations, AI services, database management, and API routing. Google Gemini API is integrated to generate AI-powered negotiation letters and personalized settlement recommendations. The application is scalable, maintainable, and designed to simplify debt management through intelligent financial analysis and secure borrower-focused features.